



## WEEKLY PROBLEM

**EXERCISE.**

Let  $M = \{f : [0, \pi] \rightarrow \mathbb{R} \mid f \text{ is twice differentiable continuous function, and } f(0) = f(\pi) = 0\}$

1. **Part A** Let  $a_0 = \frac{1}{\pi} \int_0^\pi f(x) dx$ ,  $a_n = \frac{2}{\pi} \int_0^\pi f(x) \cos(nx) dx$

Show that  $a_n \leq c/n^2$ ,  $c > 0$ ,  $n > 1$  Hence prove

$$S(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos(nx) \text{ converges uniformly}$$

2. **Part 2** Show that  $\int_0^\pi (f - S) \cos(nx) dx = 0 \forall n$  and  $f \in M$

Then conclude  $\{1, \cos x, \cos 2x, \dots\}$  after normalization forms the basis of  $M$ .



(Scan this to submit solution!)